

Sarah M. Coleman, PhD

Austin, TX → San Francisco, CA (relocating Fall/Winter 2026)

Email | LinkedIn | Website | Google Scholar | GitHub

Summary

Biochemical engineer working at the interface of experimental biology, bioprocess engineering, and data science. Background spans strain and fermentation process development, multi-omics analysis, and statistical modeling, with experience across academic (UT Austin, MIT), pharma (BMS, Takeda), and startup (BioReNuva) environments.

Education

The University of Texas at Austin, Austin, TX

Ph.D. in Chemical Engineering, Aug 2024

The University of Texas at Austin, Austin, TX

M.S. in Statistics, May 2024

Massachusetts Institute of Technology, Cambridge, MA

S.B. in Chemical-Biological Engineering, Jun 2019

Technical Skills

- **Computational Biology & Data Science:** Python (pandas, NumPy, scikit-learn), R (tidyverse), SQL, statistical modeling, machine learning, experimental design (DoE), reproducible data analysis, multi-omics analysis (RNA-seq, metabolomics, variant analysis), pathway & enrichment analysis
- **Molecular & Cellular Biology:** Yeast, bacterial and mammalian cell culture, aseptic technique, cGMP practices, CRISPR genome editing, Golden Gate assembly, strain engineering, directed/adaptive laboratory evolution, qPCR, flow cytometry, protein expression & purification, Western blotting
- **Bioprocessing & High-Throughput Experimentation:** Batch and fed-batch fermentation, bioreactor operation and control, DeltaV software, ELN software, 96-well plate screening, assay development, high-throughput screening workflows Bioinformatics & HPC
- **Workflow Engineering:** RNA-seq workflows (QC, alignment, differential expression), omics data workflows, high-performance computing (TACC), containerized pipelines (BioContainers, Docker), version-controlled workflows (Git/GitHub)
- **Analytical Chemistry & Instrumentation:** HPLC, LC-MS, GC-MS, UV-Vis, microscopy, rheology
- **Software & Reproducibility Tools:** Linux/Unix, Bash, Git/GitHub, Jupyter, Quarto, LaTeX, workflow automation

Work Experience

BioReNuva — Austin, TX

Bioprocess Development Engineer (Contract/Fixed-term) | Jun 2026 – Present

- Engaged as subject-matter expert to establish fed-batch fermentation capability for glycolipid production using attenuated *Pseudomonas aeruginosa* strains, building on prior collaboration with the Alper Lab
- Designed and implemented the company's first fed-batch bioreactor process from scratch, including feed strategy, control parameters, and run protocols
- Authored the standard operating procedure (SOP) for bioreactor runs, establishing the reference process documentation used for ongoing and future fermentation campaigns
- Defined key performance indicators (KPIs) and identified process bottlenecks for glycolipid synthesis, informing the roadmap for scale-up and continuous improvement
- Led technology transfer activities, translating process knowledge into documentation and protocols to hand off from development to manufacturing teams

The University of Texas at Austin — Austin, TX

Postdoctoral Research Fellow, Statistics and Data Sciences | Oct 2024 – Sep 2026

Research Advisor: Dr. Roger D. Peng

- Developed epidemiological and statistical models quantifying the impact of industrial air pollution mixtures on excess mortality, identifying key sector-level risk drivers
 - Modeled behavioral responses to air quality alerts to quantify pedestrian exposure-avoidance behavior under pollution events
 - Advised medical faculty on statistical methodology and experimental design in recurring data science workshops
- Lecturer, Statistics and Data Sciences* | Jan 2026 – Apr 2026
- Instructor of record for 100-student upper-division data science course, rated 4.32/5 overall by the students, a higher rating than the Department, College and University averages for Spring 2026
 - Built GitHub-based course infrastructure enabling reproducible assignments and scalable grading workflows
 - Managed 3 TAs to execute instructional, grading, and student support pipelines
- Graduate Student Fellow, Chemical Engineering* | Aug 2019 – Aug 2024
- Research Advisor: Dr. Hal S. Alper
- Integrated CFD modeling with fermentation experiments to characterize and optimize bioreactor performance in high oil density 3L bioreactor systems
 - Evolved *Y. lipolytica* to convert industrial biomass waste into the high-value polyketides triacetic acid lactone and simultaneously reduce COD (Chemical Oxygen Demand)
 - Engineered *Yarrowia lipolytica* for enhanced conversion of waste cooking oil into long-chain fatty alcohols, improving lipid hydrolysis and reducing industrial waste buildup
 - Developed a CRISPR-based genome editing system using Golden Gate assembly for targeted engineering of *Y. lipolytica*, and used this system to screen a native promoter library
 - Proposed and validated a statistical model to quantitatively evaluate the performance of biological Boolean logic gates for the first time
- MathWorks** — Remote (Natick, MA)
- MATLAB Student Ambassador* | Mar 2020 – Jul 2021
- Delivered technical workshops translating research and engineering workflows into MATLAB-based computational pipelines for academic users
 - Demonstrated applied modeling and data analysis workflows, accelerating adoption of reproducible computational methods in research environments
- Massachusetts Institute of Technology** — Cambridge, MA
- Advanced Undergraduate Researcher* | Sep 2018 – Aug 2019
- Research Advisor: Dr. Michael S. Strano
- Developed a method of nanophosphor size sorting to improve the capacitance of nanoparticles infiltrated into a light emitting plant
 - Tested the influence of nanoparticle infiltration on chlorophyll concentration and photosynthetic ability
- Undergraduate Researcher* | Sep 2016 – Aug 2018
- Research Advisor: Dr. Gregory Stephanopoulos
- Developed anaerobic assays to characterize the carbon assimilation of *Moorella thermoacetica* enzymes in the Wood-Ljungdahl pathway
- Bristol Myers Squibb** — Devens, MA
- Intern, Upstream Process Development (Biologics)* | Jun 2018 – Aug 2018
- Designed and executed metabolite-controlled fed-batch fermentation experiments to systematically evaluate monoclonal antibody production performance in mammalian cell lines
 - Analyzed upstream bioprocess data to identify conditions that improved yield stability and process robustness in biologics production
 - Supported experimental iteration cycles for process optimization in a regulated biomanufacturing environment
- Takeda Pharmaceuticals** — Cambridge, MA
- Intern, Process Development (Vaccines)* | May 2017 – Aug 2017
- Optimized mammalian cell culture conditions to improve vaccine production performance within a next-generation manufacturing system
 - Contributed to experimental design and process optimization workflows focused on improving scalability and production efficiency of biologics
- Girl's Angle Math Club** — Cambridge, MA
- Math Mentor* | Sep 2016 – Apr 2018
- Mentored students weekly in advanced mathematical reasoning with emphasis on structured problem-solving and analytical thought processes

Select Honors and Awards

- **First Place, Professional Division — Data Challenge Expo, JSM (Joint Statistical Meetings)**, 2025
Recognized for data-driven problem solving in a competitive national data science setting
- **National Science Foundation Graduate Research Fellowship (NSF GRFP)**, 2021

Prestigious national fellowship awarded for demonstrated research excellence in science

- **Provost’s Graduate Excellence Fellowship, UT Austin, 2019**
Competitive institutional fellowship awarded for outstanding incoming graduate students
- **Individual Accomplishment Citation, MIT Department of Chemical Engineering, 2018 & 2019**
Only known undergraduate to win this award two years in a row, for academic and volunteer accomplishments

Select Publications

1. Partipilo, G.*, **Coleman, S. M.***, Holwerda, A. J., Gao, Y., Mahfoud, I. E. M., Wilke, C. O., Alper, H. S., & Keitz, B. K. (2026). A Linear Mixed Effects Model for Evaluating Synthetic Gene Circuits. *ACS Synthetic Biology*. <https://doi.org/10.1021/acssynbio.6c00171>
2. **Coleman, S. M.***, Marx, R. J.*, Martinez, M. K., Silvera, A. J., Park, J., Ramanan, E., Kaown, G., Yoon, S., Xie, D., & Alper, H. S. (2024). Considerations Regarding High Oil Density Bioreactor-Scale Fermentations of *Yarrowia lipolytica* Using CFD Modeling and Experimental Validation. *Biotechnology Journal*, 19(12), e202400506. <https://doi.org/10.1002/biot.202400506>
3. **Coleman, S. M.***, Cordova, L. T.*, Lad, B. C., Ali, S. A., Ramanan, E., Collett, J. R., & Alper, H. S. (2023). Evolving tolerance of *Yarrowia lipolytica* to hydrothermal liquefaction aqueous phase waste. *Applied Microbiology and Biotechnology*. <https://doi.org/10.1007/s00253-023-12393-8>